

Effect of Real-Space Truncation on the Overcomplete Wannier Form-Factor Obstruction

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We analyze how a finite-support truncation of an overcomplete Wannier (OW) measurement mode changes the overlap form factor by which it probes a Chern band. The untruncated form factor of a band with Chern number $C \neq 0$ necessarily has zeros whose winding records the band obstruction. A real-space window produces a twisted momentum-space convolution, and the relevant question for the adaptive circuit is whether the resulting overlap $f_n^{\text{eff}}(\mathbf{k})$ retains this zero and winding. We distinguish three length scales which must not be conflated: the root-mean-square OW spread ξ_{rms} , its topological lower bound $\xi_{\text{top}} = \sqrt{2|C|} \ell$, and the asymptotic exponential tail length ξ_{exp} . For the $\alpha = 1$ model and the X -trial mode used in the code, $\xi_{\text{top}} \simeq 0.564a$, $\xi_{\text{rms}} \simeq 0.674a$, and $\xi_{\text{exp}} \simeq 1.443a$. Direct computation for the square n_{shell} mask shows that $n_{\text{shell}} = 1$ retains 99.2155% of the OW norm and preserves the displaced overlap zero with its winding. Finally, we explain why this overlap-winding statement is distinct from assigning a nonzero Fukui–Hatsugai–Suzuki Chern number to the normalized finite-support spinor itself: a smooth, periodic, nowhere-vanishing compact-support spinor is a global section and therefore has trivial Chern class.

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I. SETUP AND DEFINITIONS

Consider a single isolated band with Bloch states $|\psi_{\mathbf{k}}\rangle$ and cell-periodic part $|u_{\mathbf{k}}\rangle$, normalized so that $\langle u_{\mathbf{k}} | u_{\mathbf{k}} \rangle = 1$. Let $|\tau\rangle$ be a smooth, normalized trial spinor (we specialize to a $\mathbf{k}$ -independent trial state for clarity; the generalization is immediate) and define the form factor

$$f(\mathbf{k}) = \langle u_{\mathbf{k}} | \tau \rangle. \quad (1)$$

The coherent Wannier state centered at lattice site $\mathbf{R}$ is

$$|\chi_{\mathbf{R}}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k} \in \text{BZ}} e^{-i\mathbf{k} \cdot \mathbf{R}} f(\mathbf{k}) |\psi_{\mathbf{k}}\rangle, \quad (2)$$

which is exponentially localized since $f(\mathbf{k}) |\psi_{\mathbf{k}}\rangle = P_{\mathbf{k}} |\tau\rangle$ is smooth and periodic. The Chern number is given by the phase winding of $f(\mathbf{k})$ [1]:

$$C = -\frac{1}{2\pi i} \oint_{\text{BZ}} d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln f(\mathbf{k}). \quad (3)$$

The in-band projection of the trial state is $P_{-}(\mathbf{k}) |\tau\rangle = f(\mathbf{k}) |u_{\mathbf{k}}\rangle$, so $f(\mathbf{k})$ is also the overlap amplitude between the projected trial spinor and the lower-band eigenvector.

Connection to the overcomplete Wannier circuit. In the adaptive Markov circuit studied in the companion code, the parameter `nshell` specifies precisely the real-space support radius of the overcomplete Wannier (OW) measurement modes. An OW mode centered at $\mathbf{R}$ is constructed by projecting the trial spinor $|\tau\rangle$ with the band projector, then retaining only real-space sites $\mathbf{r}$ satisfying $|d_x(\mathbf{r}, \mathbf{R})| \leq n_{\text{shell}}$ and $|d_y(\mathbf{r}, \mathbf{R})| \leq n_{\text{shell}}$ (or the nearest-neighbor cross for $n_{\text{shell}} = 1/2$). The present analysis identifies this truncation as a real-space multiplication $g(\mathbf{r}) = g_n(\mathbf{r})$ acting on the coherent state, and the form factor of the truncated mode is precisely f_n^{eff} derived below.

II. GENERAL REAL-SPACE MODULATION

A. Definition and effective form factor

Let $g(\mathbf{r})$ be an arbitrary complex-valued function on the lattice. We define the modulated state

$$|\Phi^{(g)}\rangle = g(\hat{\mathbf{r}}) |\chi_{\mathbf{R}}\rangle, \quad (4)$$

where the multiplication is diagonal in the real-space basis: $\Phi^{(g)}(\mathbf{r}) = g(\mathbf{r}) \chi_{\mathbf{R}}(\mathbf{r})$. Writing g in terms of its Fourier transform,

$$g(\mathbf{r}) = \frac{1}{N} \sum_{\mathbf{q}} \tilde{g}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (5)$$

and substituting (2), we project onto the Bloch state $|\psi_{\mathbf{k}'}\rangle$ to define the *effective form factor*:

$$f^{(g)}(\mathbf{k}') := \sqrt{N} e^{i\mathbf{k}' \cdot \mathbf{R}} \langle \psi_{\mathbf{k}'} | \Phi^{(g)} \rangle. \quad (6)$$

Computing the inner product using $\langle \psi_{\mathbf{k}'} | \psi_{\mathbf{k}} \rangle = \delta_{\mathbf{k}', \mathbf{k}}$ on a finite lattice yields the central formula:

$$f^{(g)}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k} \in \text{BZ}} \tilde{g}(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} f(\mathbf{k}) \langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle. \quad (7)$$

Several remarks are in order.

- **Twisted convolution.** $f^{(g)}$ is a convolution of $f(\mathbf{k})$ with $\tilde{g}$, dressed by the Bloch state overlap $\langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle$ and a phase factor $e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}}$ that encodes the center of the coherent state.
- **Flat-band limit.** If $|u_{\mathbf{k}}\rangle$ is $\mathbf{k}$ -independent, the overlap is unity and the effective form factor reduces to a standard (phased) convolution:

$$f^{(g)}(\mathbf{k}')|_{\text{flat}} = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} f(\mathbf{k}). \quad (8)$$

- **Normalization independence.** Normalizing $|\Phi^{(g)}\rangle$ replaces $f^{(g)} \rightarrow f^{(g)} / \|\Phi^{(g)}\|$, an overall positive real rescaling. Since a positive constant contributes nothing to the phase winding, define the *overlap-winding index*

$$\nu_g := \frac{1}{2\pi i} \oint_{\text{BZ}} d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln f^{(g)}(\mathbf{k}). \quad (9)$$

It is independent of the normalization of $|\Phi^{(g)}\rangle$. For the unwindowed band form factor, Eq. (3) gives $\nu_0 = -C$. For a windowed state, ν_g diagnoses the winding of its overlap with the target Chern band; it is not, in general, the Chern number of the windowed real-space state.

- **Out-of-band weight.** The component of $|\Phi^{(g)}\rangle$ orthogonal to the band is $Q|\Phi^{(g)}\rangle$, where $Q = 1 - P$ is the complement projector. It is given by the same expression (7) with $\langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle$ replaced by inter-band overlaps $\langle u_{\mathbf{k}'}^{(n)} | u_{\mathbf{k}} \rangle$ for bands $n \neq 1$.

B. Topology of the modulated state

1. Vanishing of the overlap winding when $f^{(g)}$ is nowhere zero

Lemma 1. *If $f^{(g)}(\mathbf{k})$ is continuous and nowhere vanishing on T^2 , then $\nu_g = 0$.*

Proof. If $f^{(g)}(\mathbf{k}) \neq 0$ for all $\mathbf{k}$, then the map $\mathbf{k} \mapsto f^{(g)}(\mathbf{k})/|f^{(g)}(\mathbf{k})|$ defines a continuous map $T^2 \rightarrow S^1$. The total vorticity enclosed by a fundamental Brillouin-zone cell is

$$\nu_g = \frac{1}{2\pi i} \oint d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln f^{(g)}(\mathbf{k}) = \frac{1}{2\pi} \oint d\mathbf{k} \cdot \nabla_{\mathbf{k}} \arg f^{(g)}(\mathbf{k}) \quad (10)$$

and, because there are no zeros in the cell, contributions from opposite boundary edges cancel pairwise by periodicity:

$$\nu_g = \frac{1}{2\pi} \left[\int_0^1 dk_1 \partial_{k_1} \arg f^{(g)}|_{k_2=1} - \int_0^1 dk_1 \partial_{k_1} \arg f^{(g)}|_{k_2=0} \right] = 0. \quad (11)$$

Thus there is no net zero vorticity enclosed by the Brillouin-zone torus. This statement does not forbid independent noncontractible S^1 windings of a nowhere-vanishing complex scalar along individual cycles of T^2 ; those are not the Chern-obstruction index considered here. $\square$

2. Preservation of zeros: narrow $\tilde{g}$

Lemma 2. *Suppose $f(\mathbf{k})$ has a zero of order $|C|$ and winding $-C$ at $\mathbf{k}_0$, with $f(\mathbf{k}_0 + \mathbf{q}) \approx A_0(q_x - i \operatorname{sgn}(C) q_y)^{|C|}$ for $|\mathbf{q}| \leq \delta k$. Let $\tilde{g}$ satisfy $|\tilde{g}(\mathbf{q})| \leq \epsilon$ for $|\mathbf{q}| > \Lambda$ with $\Lambda \ll \delta k$. Then $f^{(g)}$ has a zero near $\mathbf{k}_0$ with the same winding, provided ϵ is sufficiently small.*

Proof. We temporarily work in the flat-band limit; the general case follows since the overlap deviates from $\delta_{\mathbf{k}', \mathbf{k}}$ by $O(|\mathbf{k}' - \mathbf{k}|) = O(\Lambda)$. In this limit, $f^{(g)}(\mathbf{k}')$ is a phased convolution

$$f^{(g)}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} f(\mathbf{k}). \quad (12)$$

Define $h(\mathbf{k}) = \tilde{g}(\mathbf{k}) e^{-i\mathbf{k} \cdot \mathbf{R}}$; then $f^{(g)} = (1/N)(h * f)$.

Consider a circle γ_ρ of radius ρ around $\mathbf{k}_0$ with $\Lambda \ll \rho \ll \delta k$. On γ_ρ , $|f(\mathbf{k}_0 + \mathbf{q})| \sim |A_0| \rho^{|C|}$. Splitting the convolution correction into $|\mathbf{q}| \leq \Lambda$ and $|\mathbf{q}| > \Lambda$:

$$|f^{(g)}(\mathbf{k}) - f(\mathbf{k})| \lesssim |A_0| |C| \rho^{|C|-1} \Lambda + \epsilon \|f\|_\infty. \quad (13)$$

For $\rho \gg \Lambda$ and ϵ small, both terms are $\ll |A_0| \rho^{|C|} \sim |f(\mathbf{k})|$ on γ_ρ . By Rouché's theorem, $f^{(g)}$ and f have the same zeros counted with winding inside γ_ρ , giving $\nu_g = \nu_0 = -C$. $\square$

3. Destruction of zeros: broad $\tilde{g}$

Lemma 3. Suppose $f(\mathbf{k})$ has a zero at $\mathbf{k}_0$ with $|f(\mathbf{k}_0 + \mathbf{q})| \lesssim |A_0| |\mathbf{q}|^{|C|}$ for $|\mathbf{q}| \leq \delta k$, and $|f(\mathbf{k})| \geq f_{\min} > 0$ for $|\mathbf{k} - \mathbf{k}_0| \geq \delta k$. Let $|h(\mathbf{q})| = |\tilde{g}(\mathbf{q}) e^{-i\mathbf{q} \cdot \mathbf{R}}|$ be smooth with $|h(\mathbf{q})| \geq h_0 > 0$ for $|\mathbf{q}| \leq \Lambda$ and $\Lambda \gg \delta k$. Then $f^{(g)}(\mathbf{k}_0) \neq 0$ generically.

Proof. Evaluate (26) in the flat-band limit, decomposing into the zero region $Z = \{|\mathbf{q}| \leq \delta k\}$ and its complement:

$$f^{(g)}(\mathbf{k}_0) = \underbrace{\frac{1}{N} \sum_{|\mathbf{q}| \leq \delta k} h(\mathbf{q}) f(\mathbf{k}_0 + \mathbf{q})}_{I_Z} + \underbrace{\frac{1}{N} \sum_{|\mathbf{q}| > \delta k} h(\mathbf{q}) f(\mathbf{k}_0 + \mathbf{q})}_{I_{Z^c}}. \quad (14)$$

One finds $|I_Z| = O(\sigma^2 (\delta k)^{|C|+2})$ while $|I_{Z^c}| = O(\sigma^2 f_{\min})$ generically (i.e., absent fine-tuned phase cancellation in I_{Z^c}). Since $(\delta k)^{|C|+2}/|\mathbf{b}|^2 \ll 1$, we have $|I_Z|/|I_{Z^c}| \rightarrow 0$ and $f^{(g)}(\mathbf{k}_0) \neq 0$. $\square$

Remark. The word “generically” cannot be removed: one can always construct pathological g (e.g., $g = 0$) producing fine-tuned cancellations. The content of Lemma 3 is that for g drawn from any reasonable parametric family, the zero is filled once the kernel width exceeds δk , except on a measure-zero set of parameters.

4. Three distinct localization scales

The spread entering the quantum-geometric bound is the second moment of the unwinded OW coherent state,

$$\xi_{\text{rms}}^2 := \langle \chi_{\mathbf{R}=0} | \mathbf{r}^2 | \chi_{\mathbf{R}=0} \rangle. \quad (15)$$

The Li–Dong–Ledwith–Khalaf bound states

$$\xi_{\text{rms}} \geq \xi_{\text{top}}, \quad \xi_{\text{top}} := \sqrt{2|C|} \ell, \quad 2\pi \ell^2 = A_{\text{uc}}, \quad \ell^2 = \frac{2\pi}{A_{\text{BZ}}}, \quad (16)$$

where $A_{\text{uc}} A_{\text{BZ}} = (2\pi)^2$. For a square lattice with $A_{\text{uc}} = a^2$ and $|C| = 1$, this lower bound is

$$\xi_{\text{top}} = \frac{a}{\sqrt{\pi}} \simeq 0.564 a. \quad (17)$$

The zero-resolution momentum scale may be estimated parametrically by $\delta k = O(\xi_{\text{rms}}^{-1})$, so a window broad in real space compared with the actual OW spread gives a narrow convolution kernel. This is a scale estimate, not a theorem that a window of radius ξ_{top} preserves a particular zero.

A second, independent length is the asymptotic tail-decay scale ξ_{exp} , defined from the large-distance envelope

$$\|\chi_0(\mathbf{r})\| \sim e^{-|\mathbf{r}|/\xi_{\text{exp}}} \quad (|\mathbf{r}| \rightarrow \infty), \quad (18)$$

up to algebraic prefactors. It controls small discarded tails rather than the RMS weight or the survival of the overlap-winding singularity. In particular, $\xi_{\text{exp}} > \xi_{\text{rms}}$ does not imply that a compact window whose radius is below ξ_{exp} destroys the form-factor zero.

III. SPECIALIZATION: BOX FUNCTION OF RADIUS σ

We now specialize to the circular box (indicator) function

$$g(\mathbf{r}) = B_\sigma(\mathbf{r} - \mathbf{R}') := \begin{cases} 1 & |\mathbf{r} - \mathbf{R}'| \leq \sigma \\ 0 & |\mathbf{r} - \mathbf{R}'| > \sigma \end{cases}, \quad (19)$$

whose Fourier transform is

$$\tilde{g}(\mathbf{q}) = \tilde{B}_\sigma(\mathbf{q}) e^{-i\mathbf{q} \cdot \mathbf{R}'}, \quad \tilde{B}_\sigma(\mathbf{q}) = \frac{2\pi\sigma}{|\mathbf{q}|} J_1(|\mathbf{q}|\sigma), \quad (20)$$

with J_1 the Bessel function of the first kind. The central lobe has width $\Delta k \sim 1/\sigma$, but $\tilde{B}_\sigma$ has oscillating power-law tails $\sim |\mathbf{q}|^{-3/2}$ at large $|\mathbf{q}|$. Substituting into (7) and defining $\mathbf{\Delta} = \mathbf{R}' - \mathbf{R}$:

$$f_\sigma(\mathbf{k}') = e^{-i\mathbf{k}' \cdot \mathbf{\Delta}} \frac{1}{N} \sum_{\mathbf{k}} \tilde{B}_\sigma(\mathbf{k}' - \mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{\Delta}} f(\mathbf{k}) \langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle. \quad (21)$$

The overall phase $e^{-i\mathbf{k}' \cdot \mathbf{\Delta}}$ has zero winding and drops out of ν_σ . When $\mathbf{\Delta} = 0$ (box centered on the coherent state), (21) reduces to the twisted convolution of f with $\tilde{B}_\sigma$, dressed by the Bloch overlap.

Power-law tails of $\tilde{B}_\sigma$

Unlike a smooth window (Gaussian, etc.) whose Fourier transform is rapidly decaying, $\tilde{B}_\sigma(\mathbf{q})$ has power-law tails $\sim |\mathbf{q}|^{-3/2} \cos(|\mathbf{q}|\sigma - 3\pi/4)$. This means the box convolution mixes momenta over *all* scales, albeit with decreasing weight. The dominant contribution still comes from $|\mathbf{q}| \lesssim 1/\sigma$, so comparison with ξ_{rms} remains a useful parametric estimate. However, the tails imply that the crossover between preserved and removed overlap winding is less sharp than for a smooth window.

IV. GRADIENT EXPANSION FOR $\sigma \gg a$

Restricting to the centered case $\mathbf{R}' = \mathbf{R}$ and expanding (21) for $\sigma \gg a$ (lattice spacing), we use the moments of the box kernel ($\int \mathbf{q} \tilde{B}_\sigma(\mathbf{q}) = 0$ by radial symmetry; $\int q_\mu q_\nu \tilde{B}_\sigma(\mathbf{q}) = \frac{\delta_{\mu\nu}}{4} \sigma^{-2}$) and expand the Bloch overlap to second order:

$$\langle u_{\mathbf{k}'} | u_{\mathbf{k}'+\mathbf{q}} \rangle = 1 - iA_\mu(\mathbf{k}')q_\mu - \frac{1}{2} \left[g_{\mu\nu}(\mathbf{k}') + i\frac{1}{2}\epsilon_{\mu\nu}\Omega(\mathbf{k}') \right] q_\mu q_\nu + O(q^3), \quad (22)$$

where $A_\mu = i \langle u_{\mathbf{k}} | \partial_{k_\mu} u_{\mathbf{k}} \rangle$ is the Berry connection, $g_{\mu\nu} = \text{Re} \langle \partial_{k_\mu} u_{\mathbf{k}} | Q_{\mathbf{k}} \partial_{k_\nu} u_{\mathbf{k}} \rangle$ is the quantum metric, and Ω is the Berry curvature. The result is

$$f_\sigma(\mathbf{k}) = (f * \tilde{B}_\sigma)(\mathbf{k}) - iA_\mu(\mathbf{k}) (\partial_{k_\mu} f * \tilde{B}_\sigma)(\mathbf{k}) + O(\sigma^{-2}). \quad (23)$$

The leading term is a scalar convolution; the first correction involves the Berry connection and is suppressed by $O(1/\sigma)$.

V. ZEROS OF f_σ AND THE OVERLAP-WINDING INDEX

We specialize to the centered case $\mathbf{R}' = \mathbf{R}$, so that

$$f_\sigma(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{B}_\sigma(\mathbf{k}' - \mathbf{k}) f(\mathbf{k}) \langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle. \quad (24)$$

Following Ref. [1], the zeros of $f(\mathbf{k})$ can be arranged in the single-zero (SZ) gauge where there is a single zero at $\mathbf{k}_0$ with

$$f(\mathbf{k}_0 + \mathbf{q}) = A_0(q_x - i \text{sgn}(C) q_y)^{|C|} + R(\mathbf{q}), \quad (25)$$

where $A_0 \neq 0$ and $R(\mathbf{q}) = O(|\mathbf{q}|^{|C|+1})$.

A. Case 1: $\sigma \gg \xi_{\text{rms}}$ — preservation of zeros

Proposition (Case 1). *Let $\sigma \gg \xi_{\text{rms}}$ and $\tilde{B}_\sigma$ be the Fourier transform of a circular box of radius σ . If the convolved overlap is a sufficiently small perturbation of f on a loop surrounding $\mathbf{k}_0$, then f_σ has a zero near $\mathbf{k}_0$ with winding $-C$, so that $\nu_\sigma = -C$.*

Proof. Step 1: Decomposition. Write (24) at $\mathbf{k}' = \mathbf{k}_0$ as

$$f_\sigma(\mathbf{k}_0) = \frac{1}{N} \sum_{\mathbf{q}} \tilde{B}_\sigma(\mathbf{q}) f(\mathbf{k}_0 + \mathbf{q}) \langle u_{\mathbf{k}_0} | u_{\mathbf{k}_0 + \mathbf{q}} \rangle, \quad (26)$$

decomposed into near ($|\mathbf{q}| \leq \rho$) and far ($|\mathbf{q}| > \rho$) regions with $1/\sigma \ll \rho \ll \delta k$.

Step 2: Leading near-region harmonic. Using the leading zero structure (25) and passing to polar coordinates:

$$I_{\text{near}} \propto \int_0^\rho q dq \tilde{B}_\sigma(q) q^{|C|} \underbrace{\int_0^{2\pi} \frac{d\phi_q}{2\pi} e^{-i|C|\phi_q}}_{=0 \text{ for } |C| \geq 1} = 0. \quad (27)$$

The displayed leading harmonic integrates to zero against a radial kernel. Subleading terms from $R(\mathbf{q})$ and from the Bloch-overlap factor need not vanish exactly; rather, in the narrow-kernel regime they are perturbative and can shift the zero away from $\mathbf{k}_0$ without changing its winding.

Step 3: Winding stability. For $|\mathbf{q}| > \rho \gg 1/\sigma$, the kernel is in its asymptotic tail regime $|\tilde{B}_\sigma(\mathbf{q})| \lesssim C_B \sigma^{1/2} |\mathbf{q}|^{-3/2}$. Applying Rouché's theorem on a circle γ_ρ of radius ρ around $\mathbf{k}_0$ whenever the total correction is smaller than $|f|$ on that circle establishes that f_σ has the same number of zeros with the same winding as f inside γ_ρ , hence $\nu_\sigma = -C$. $\square$

B. Case 2: $\sigma \ll \xi_{\text{rms}}$ — destruction of zeros

Proposition (Case 2). *Let $\sigma \ll \xi_{\text{rms}}$. Then generically $f_\sigma(\mathbf{k}_0) \neq 0$; if no replacement zero is generated elsewhere on T^2 , the overlap winding is $\nu_\sigma = 0$.*

Proof. Direct application of Lemma 3. Since $\sigma \ll \xi_{\text{rms}}$ places the central lobe of $\tilde{B}_\sigma$ broad compared with the zero-resolution scale, substantial kernel weight enters from outside the zero region. The ratio $|I_Z|/|I_{Z^c}| = O((\delta k)^{|C|+2}/|\mathbf{b}|^2) \ll 1$ forces $f_\sigma(\mathbf{k}_0) \approx I_{Z^c} \neq 0$ absent symmetry-enforced cancellation. If f_σ is nowhere zero after this rearrangement, Lemma 1 gives $\nu_\sigma = 0$. $\square$

C. Crossover and the topological bound

The crossover between Cases 1 and 2 is estimated by comparing the window size with the actual OW spread, so that the central-lobe width $\Delta k = 1/\sigma$ resolves the form-factor zero. The geometric lower bound (16) implies only the parametric lower scale

$$\boxed{\sigma_{\text{crit}} = O(\xi_{\text{rms}}), \quad \xi_{\text{rms}} \geq \xi_{\text{top}} = \sqrt{2|C|} \ell.} \quad (28)$$

For σ large compared with the relevant spread, overlap winding is stable; for aggressive truncation it may be removed or reorganized. The inequality constrains the possible OW spread, but does not certify any particular discrete window. In particular, the survival of the $n_{\text{shell}} = 1$ zero must be and is checked directly below.

VI. MULTIBAND GENERALIZATION

A. Setup

Let $|\psi_{n,\mathbf{k}}\rangle$, $n = 1, \dots, M$, be the Bloch states of M isolated bands with cell-periodic parts $|u_{n,\mathbf{k}}\rangle$. Construct M trial states $|\tau_m\rangle$ and define the $M \times M$ overlap matrix

$$F_{nm}(\mathbf{k}) = \langle u_{n,\mathbf{k}} | \tau_m \rangle. \quad (29)$$

The total Chern number of the M -band subspace is [1]

$$C = -\frac{1}{2\pi i} \oint_{\text{BZ}} d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln \det F(\mathbf{k}). \quad (30)$$

The multiband coherent states are

$$|\chi_{m,\mathbf{R}}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \sum_{n=1}^M F_{nm}(\mathbf{k}) |\psi_{n,\mathbf{k}}\rangle. \quad (31)$$

B. Effective overlap matrix and its topology

Define modulated states $|\Phi_m^{(g)}\rangle = g(\hat{\mathbf{r}}) |\chi_{m,\mathbf{R}}\rangle$ and the *effective overlap matrix*:

$$F_{nm}^{(g)}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \sum_{n'=1}^M \tilde{g}(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k})\cdot\mathbf{R}} F_{n'm}(\mathbf{k}) \langle u_{n,\mathbf{k}'} | u_{n',\mathbf{k}} \rangle. \quad (32)$$

This is the matrix generalization of (7). The determinant overlap-winding index of the modulated states relative to the target subspace is

$$\nu_g^{(M)} = \frac{1}{2\pi i} \oint_{\text{BZ}} d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln \det F^{(g)}(\mathbf{k}). \quad (33)$$

If the modulated vectors are projected back into and form a nonsingular frame for the target subspace, this winding reproduces the subspace obstruction. For general real-space-windowed vectors with out-of-band weight it is an overlap diagnostic rather than an independent Chern invariant. An important structural difference from the single-band case is that $\det F^{(g)}$ is a polynomial in the convolved matrix elements, so zeros can in principle be created or annihilated in pairs by the modulation even at intermediate scales.

In the single-zero gauge, $\det F(\mathbf{k}_0 + \mathbf{q}) \approx A_0(q_x - i \operatorname{sgn}(C) q_y)^{|C|}$. The narrow-kernel continuity argument applies to $\det F^{(g)}$ when $\sigma \gg \xi_{\text{rms}}$, with the same qualification that the zero may shift while its determinant-overlap winding remains stable.

The critical modulation scale for the multiband case remains

$$\sigma_{\text{crit}} = O(\xi_{\text{rms}}), \quad \xi_{\text{rms}} \geq \sqrt{2|C|} \ell, \quad (34)$$

where C is now the total Chern number of the M -band subspace.

VII. COMPUTATIONAL EXAMPLE: SQUARE n_{shell} MASK

The numerical notebook accompanying this note studies the two-band lattice Chern-insulator Hamiltonian

$$h(\mathbf{k}) = \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad \mathbf{n}(\mathbf{k}) = (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \quad \hat{\mathbf{n}} = \frac{\mathbf{n}}{|\mathbf{n}|}, \quad (35)$$

with lower-band projector $P_{-}(\mathbf{k}) = (1 - \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma})/2$. For a normalized trial spinor $|\tau\rangle$, the form-factor amplitude is

$$|f(\mathbf{k})|^2 = \langle \tau | P_{-}(\mathbf{k}) | \tau \rangle = \frac{1 - \hat{\mathbf{n}}(\mathbf{k}) \cdot \mathbf{s}_{\tau}}{2}, \quad \mathbf{s}_{\tau} = \langle \tau | \boldsymbol{\sigma} | \tau \rangle. \quad (36)$$

For the X trial spinor $|\tau_X\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ with $\mathbf{s}_{\tau_X} = (1, 0, 0)$:

$$|f_X(\mathbf{k})|^2 = \frac{1}{2} \left[1 - \frac{\sin k_x}{\sqrt{\sin^2 k_x + \sin^2 k_y + (\alpha - \cos k_x - \cos k_y)^2}} \right]. \quad (37)$$

A true zero requires $\hat{n}_x = 1$, i.e., $\sin k_y = 0$, $\alpha - \cos k_x - \cos k_y = 0$, $\sin k_x > 0$. For $\alpha = 1$ this gives

$$\mathbf{k}_0/\pi = (0.5, 0). \quad (38)$$

Instead of the circular disk window used for analytic estimates, the project code implements the square real-space mask

$$g_n(\mathbf{r}) = \begin{cases} 1, & |d_x(\mathbf{r}, \mathbf{R})| \leq n, \quad |d_y(\mathbf{r}, \mathbf{R})| \leq n, \\ 0, & \text{otherwise,} \end{cases} \quad (39)$$

where d_x, d_y are the periodic lattice distances from the window center $\mathbf{R}$. For the special case $n_{\text{shell}} = 1/2$, only the five-site cross is kept: $\{(0,0), (\pm 1,0), (0,\pm 1)\}$. In the thermodynamic, centered case, the Fourier transform of the square mask factorizes into Dirichlet kernels,

$$\tilde{g}_n(\mathbf{q}) = e^{-i\mathbf{q}\cdot\mathbf{R}} D_n(q_x) D_n(q_y), \quad D_n(q) = \sum_{m=-n}^n e^{-iqm} = \frac{\sin[(n + \frac{1}{2})q]}{\sin(q/2)}, \quad (40)$$

and the finite-grid computation evaluates

$$f_n^{\text{eff}}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}_n(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k})\cdot\mathbf{R}} f(\mathbf{k}) \langle u_-(\mathbf{k}') | u_-(\mathbf{k}) \rangle. \quad (41)$$

Figure 1 shows the exact form factor and its two spinor components on a 50×50 momentum grid. The true zero at $\mathbf{k}/\pi = (0.5, 0)$ lies between sampled momenta; the nearest sampled minimum occurs at $\mathbf{k}/\pi = (0.48, 0)$ with $|f| \simeq 3.14 \times 10^{-2}$. Figure 2 shows the real-space support of the truncation for representative values of n_{shell} .

A. Localization scales and retained norm at $\alpha = 1$

To compare the finite window with the actual OW mode, we use the same lower-band projector and X -trial spinor as in the circuit implementation and set the lattice spacing to $a = 1$ for the reported numbers. Direct Fourier evaluation of $P_-(\mathbf{k}) |\tau_X\rangle$ gives

$$\xi_{\text{rms}}^2 = 0.453631 a^2, \quad \xi_{\text{rms}} = 0.673521 a. \quad (42)$$

The nearest complex-momentum singularity of the flattened projector controls the long-distance envelope. At $\alpha = 1$,

$$|\mathbf{n}(\mathbf{k})|^2 = 3 - 2 \cos k_x - 2 \cos k_y + 2 \cos k_x \cos k_y. \quad (43)$$

For transverse decay, setting $k_x = i\kappa$ and taking the nearest real value $k_y = \pi$ gives

$$5 - 4 \cosh \kappa = 0, \quad \kappa = \text{arcosh} \frac{5}{4} = \log 2, \quad \xi_{\text{exp}} = \frac{a}{\log 2} = 1.442695 a. \quad (44)$$

The three lengths are therefore

scale	value at $\alpha = 1$	meaning
ξ_{top}	$0.564190 a$	topological lower bound on RMS spread
ξ_{rms}	$0.673521 a$	actual OW second-moment radius
ξ_{exp}	$1.442695 a$	asymptotic exponential tail length

(45)

These quantities answer different questions. In particular, the comparatively long asymptotic tail does not imply that most probability weight lies outside a 3×3 window. Applied to the normalized untruncated OW mode, the square masks retain

n_{shell}	support	$\sum_{\mathbf{r} \in \text{support}} \ \chi_0(\mathbf{r})\ ^2$
1	3×3	0.992155
2	5×5	0.999137

(46)

of its norm. Thus $n_{\text{shell}} = 1$ removes a small long-distance tail; whether it retains the topological overlap obstruction is determined by the form-factor zero below, not by comparing the cutoff radius with ξ_{exp} alone.

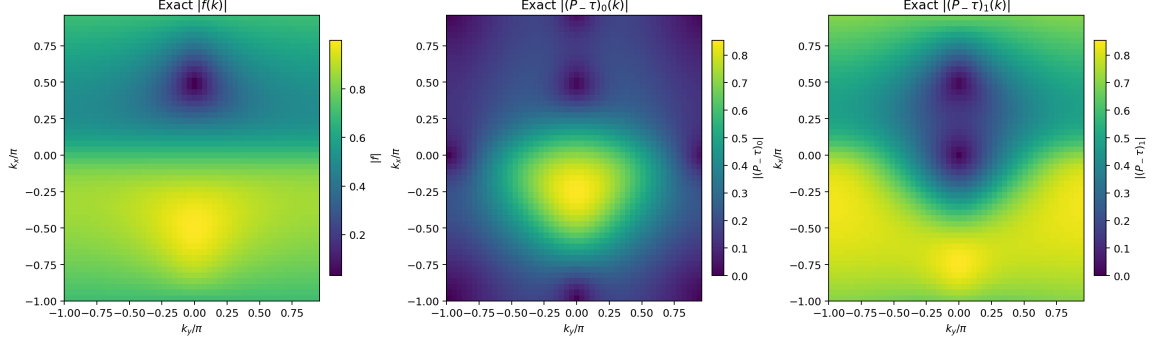


FIG. 1. Exact lower-band X -trial form factor for the two-band model at $\alpha = 1$ on a 50×50 momentum grid. Left: $|f(k)|$, with the true continuum zero at $\mathbf{k}/\pi = (0.5, 0)$ between sampled momenta. Center and right: the two components of the projected trial state $P_-(\mathbf{k})|\tau_X\rangle$.

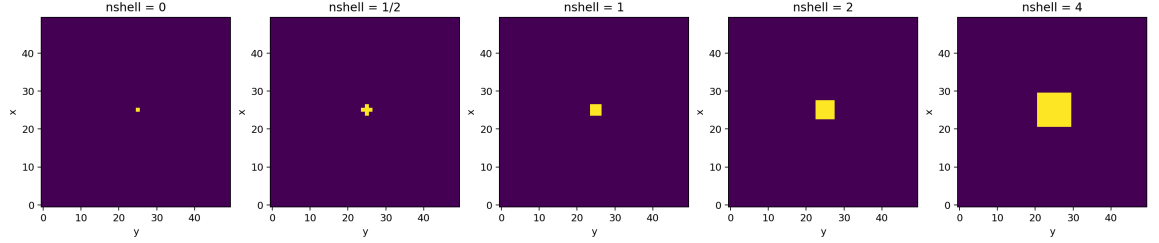


FIG. 2. Real-space square masks used for the finite- n_{shell} calculation. The $n_{\text{shell}} = 0$ case is a single-site mask, hence $\tilde{g}_0(\mathbf{q})$ is flat in momentum space and mixes the whole BZ. Increasing n_{shell} concentrates the Fourier kernel near $\mathbf{q} = 0$, enabling a sharper effective form factor.

B. The $n_{\text{shell}} = 0$ limit

For $n_{\text{shell}} = 0$, the mask reduces to the single site at the origin:

$$g_0(\mathbf{r}) = \delta_{\mathbf{r}, \mathbf{R}}, \quad \tilde{g}_0(\mathbf{q}) = 1 \quad (\text{centered case}). \quad (47)$$

Substituting into the effective form factor:

$$f_0^{\text{eff}}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} f(\mathbf{k}) \langle u_-(\mathbf{k}') | u_-(\mathbf{k}) \rangle. \quad (48)$$

This is a full BZ average of $f(\mathbf{k})$ weighted by the Bloch overlap, not a local evaluation of f at $\mathbf{k}'$. The original zero of f does not force a zero of f_0^{eff} .

In the continuum limit, only the onsite Fourier coefficient

$$C_0 = \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} P_-(\mathbf{k}) \quad (49)$$

appears. By the square-lattice symmetries at $\alpha = 1$, $C_0 = \text{diag}(a, 1 - a)$ with $a \simeq 0.254415$, and on the $k_y = 0$ line:

$$f_0^{\text{eff}}(k_x, 0) = \frac{1}{\sqrt{2}} \left[a \cos \frac{k_x}{2} - (1 - a) \sin \frac{k_x}{2} \right]. \quad (50)$$

This has an accidental zero at $\tan(k_x/2) = a/(1 - a)$, i.e.,

$$\mathbf{k}_{*,0}/\pi \simeq (0.2093, 0), \quad (51)$$

which is the continuum counterpart of the shallow 50×50 dip near $k_x/\pi = 0.20$ ($|f_0^{\text{eff}}| \simeq 8.18 \times 10^{-3}$). This is not the topological zero at $k_x/\pi = 0.5$: it is an accidental zero of the BZ-averaged object, and it carries the correct winding (see below) only because a zero must exist somewhere on T^2 for any state with $C = 1$ topology.

C. Explicit checks for $n_{\text{shell}} = 1$ and 2

For $n_{\text{shell}} = 1$ (the 3×3 square mask), $D_1(q) = 1 + 2 \cos q$ and $\tilde{g}_1(q_x, q_y) = D_1(q_x)D_1(q_y)$. At $\mathbf{k}_0 = (\pi/2, 0)$, where $\hat{\mathbf{n}}(\mathbf{k}_0) = (1, 0, 0)$ and the lower-band spinor is the $-\sigma_x$ eigenstate, one computes

$$f_1^{\text{eff}}(\mathbf{k}_0) = -\frac{1}{2N} \sum_{\mathbf{k}} (1 + 2 \cos k_y) \frac{1 - \cos k_x - \cos k_y}{|\mathbf{n}(\mathbf{k})|} \simeq -1.21 \times 10^{-2}, \quad (52)$$

so the original zero is displaced from $\mathbf{k}_0$. On the symmetry line $k_y = 0$, the shifted continuum zero occurs at

$$\mathbf{k}_{*,1}/\pi \simeq (0.4922, 0). \quad (53)$$

For $n_{\text{shell}} = 2$ (the 5×5 mask), $D_2(q) = 1 + 2 \cos q + 2 \cos 2q$ and similarly

$$f_2^{\text{eff}}(\mathbf{k}_0) \simeq -1.17 \times 10^{-2}, \quad \mathbf{k}_{*,2}/\pi \simeq (0.4925, 0). \quad (54)$$

In both cases the zero is shifted rather than removed. The shift converges to $k_x/\pi = 0.5$ as $n_{\text{shell}} \rightarrow \infty$.

D. Persistence of the zero for sufficiently large finite shells

For nonzero n_{shell} , the effective form factor near the original zero at $\mathbf{k}_0 = (\pi/2, 0)$ can be expanded as follows. Write $\mathbf{k}' = \mathbf{k}_0 + \mathbf{p}$ and $\mathbf{k} = \mathbf{k}' + \mathbf{q}$. In the centered case $\mathbf{R} = 0$:

$$\begin{aligned} f_n^{\text{eff}}(\mathbf{k}_0 + \mathbf{p}) &= \frac{A}{N} \sum_{\mathbf{q}} \tilde{g}_n(-\mathbf{q}) [(p_x + q_x) - i(p_y + q_y)] + O(\Delta k_n^2) \\ &= A G_0 (p_x - ip_y) + \underbrace{\frac{A}{N} \sum_{\mathbf{q}} \tilde{g}_n(-\mathbf{q}) q_x}_{=0} - \underbrace{\frac{iA}{N} \sum_{\mathbf{q}} \tilde{g}_n(-\mathbf{q}) q_y}_{=0} + O(\Delta k_n^2), \end{aligned} \quad (55)$$

where $G_0 = (1/N) \sum_{\mathbf{q}} \tilde{g}_n(-\mathbf{q}) = g_n(\mathbf{0}) = 1$ and $\Delta k_n \sim 1/(2n+1)$. The first moments vanish because the square mask is inversion-symmetric, $g_n(\mathbf{r}) = g_n(-\mathbf{r})$, making $\tilde{g}_n(\mathbf{q})$ real and even. Therefore,

$$f_n^{\text{eff}}(\mathbf{k}_0 + \mathbf{p}) = A (p_x - ip_y) + O(\Delta k_n^2, p^2, p \Delta k_n), \quad (56)$$

and the leading winding structure is identical to the original zero. By a Rouché-continuity argument, for sufficiently large n_{shell} there is a zero of f_n^{eff} near $\mathbf{k}_0$ with the same phase winding.

This is in sharp contrast to the $n_{\text{shell}} = 0$ case, where $\tilde{g}_0$ is completely broad and the small parameter Δk_n does not exist.

E. Continuum- k effective form factor cuts

To go beyond the finite 50×50 sampling grid, we compute the effective spinor state directly from the projector's Fourier coefficients,

$$|\psi_n(\mathbf{k})\rangle = \sum_{\mathbf{r} \in \text{supp}(g_n)} e^{-i\mathbf{k} \cdot \mathbf{r}} C_{\mathbf{r}} |\tau\rangle, \quad C_{\mathbf{r}} = \int_{\text{BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\mathbf{q} \cdot \mathbf{r}} P_{-}(\mathbf{q}), \quad (57)$$

evaluated on a dense 181×181 momentum grid. The scalar effective form factor is then $f_n^{\text{eff}}(\mathbf{k}) = \langle u_{-}(\mathbf{k}) | \psi_n(\mathbf{k}) \rangle$, where the lower-band eigenspinor $|u_{-}(\mathbf{k})\rangle$ is determined at each $\mathbf{k}$ from a local eigenvector decomposition. This formulation avoids the discrete-grid Fourier artifact of the finite- N computation and yields the true continuum zeros.

Figure 3 shows the resulting $k_y = 0$ amplitude cuts $|f_n^{\text{eff}}(k_x, 0)|$ on the dense continuum grid, and Figure 4 shows the same cuts with the shifted zero locations marked explicitly. For $n_{\text{shell}} \geq 1$, the zeros converge toward $k_x/\pi = 0.5$ as n_{shell} increases; for $n_{\text{shell}} = 0$, the zero is far displaced to $k_x/\pi \simeq 0.209$.

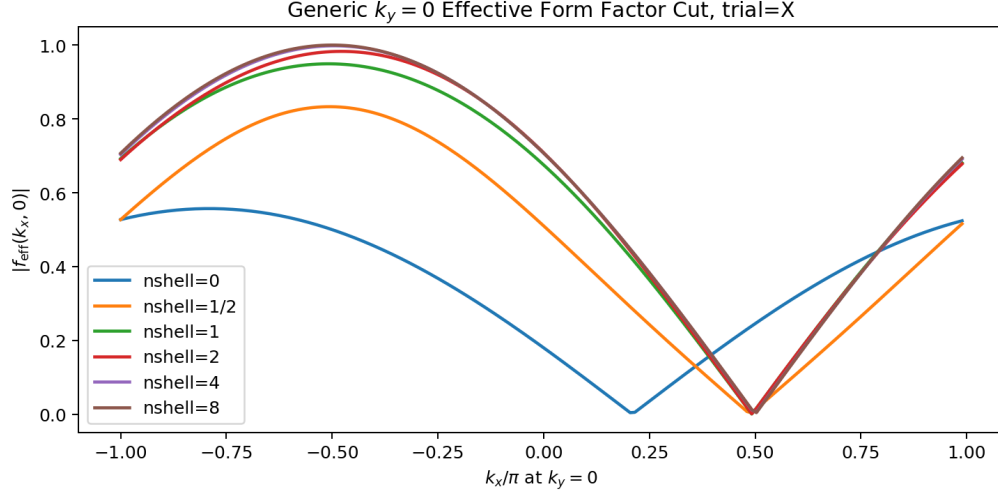


FIG. 3. Continuum- $\mathbf{k}$ cuts of $|f_n^{\text{eff}}(k_x, 0)|$ at $k_y = 0$ on a 181×181 grid, computed via the projector Fourier coefficients (57). Each curve corresponds to a different n_{shell} value. The minimum for $n_{\text{shell}} = 0$ is located far from the original topological zero at $k_x/\pi = 0.5$, while the minima for $n_{\text{shell}} \geq 1$ approach $k_x/\pi = 0.5$ from below.

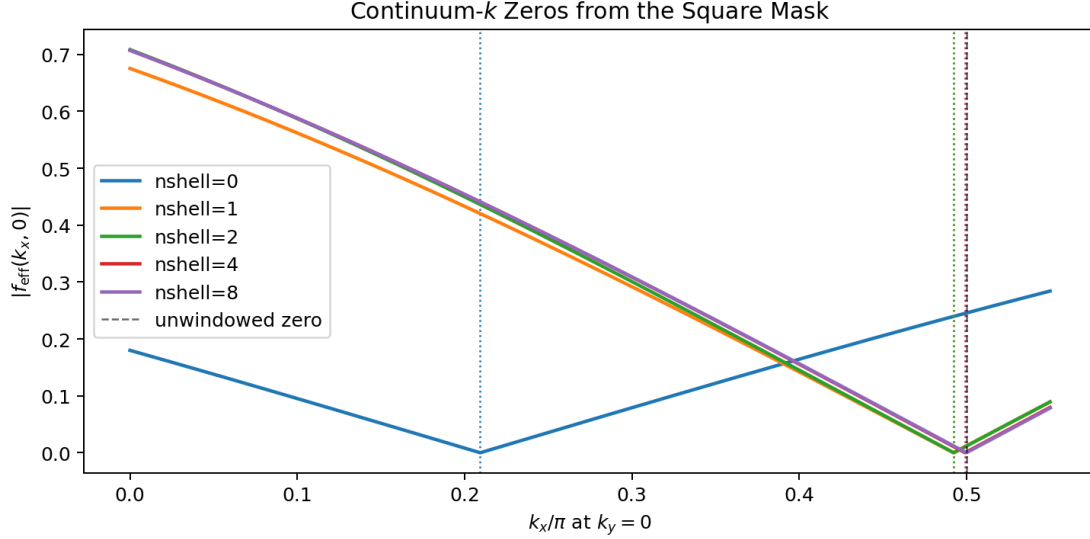


FIG. 4. Continuum- $\mathbf{k}$ cuts of the finite-shell effective form factor on $k_y = 0$. The dashed vertical line is the unwound zero at $k_x/\pi = 0.5$. For $n_{\text{shell}} = 0$, the dotted line near $k_x/\pi \simeq 0.209$ is an accidental zero of the fully BZ-averaged onsite truncation. For $n_{\text{shell}} = 1, 2, 4, 8$, the dotted lines show the shifted descendants of the unwound zero, which converge monotonically toward $k_x/\pi = 0.5$.

F. Effective form factor maps and the approach to the unwound limit

The full BZ dependence of $|f_n^{\text{eff}}(\mathbf{k})|$ is shown in Figures 5 and 6. As n_{shell} increases, the effective form factor approaches the unwound result: the amplitude maximum rises toward unity (see the printed values $\max |f_n^{\text{eff}}| = 0.556, 0.988, 0.949, 0.983, 0.998, 0.999$ for $n = 0, 1/2, 1, 2, 4, 8$) and the zero approaches the continuum location $\mathbf{k}/\pi = (0.5, 0)$.

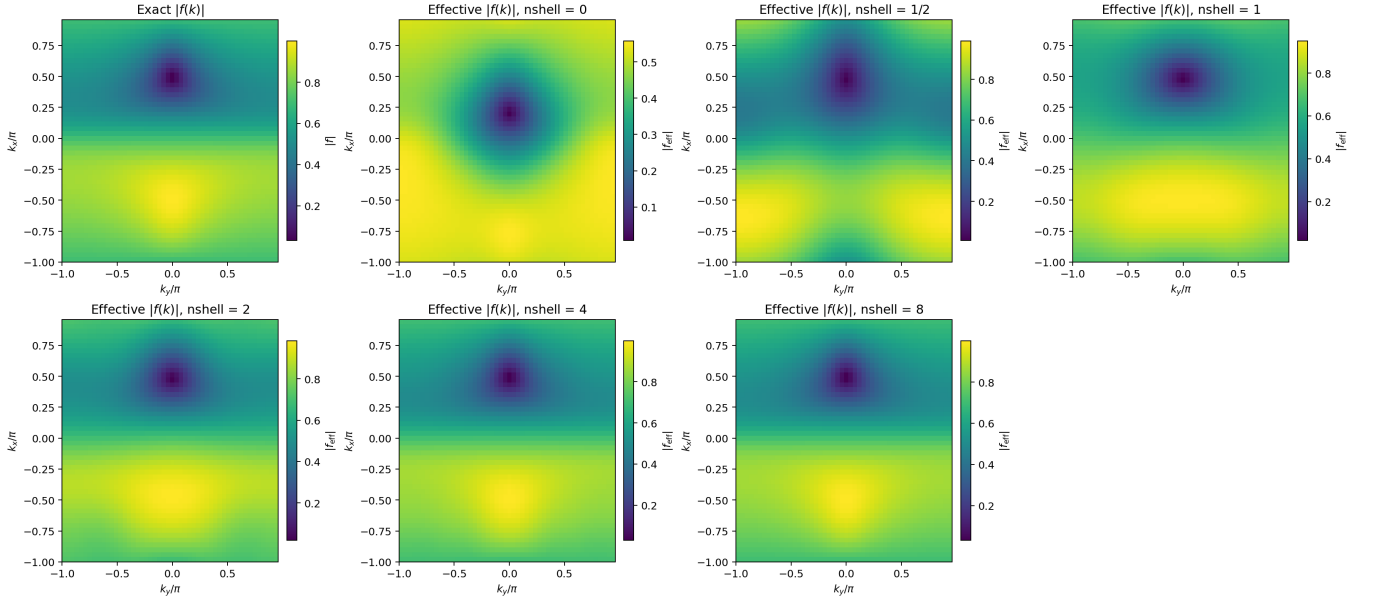


FIG. 5. Momentum-space amplitude $|f_n^{\text{eff}}(\mathbf{k})|$ from the square real-space mask, for the exact unwinded form factor (top left) and for $n_{\text{shell}} = 0, 1/2, 1, 2, 4, 8$. Increasing n_{shell} concentrates the effective form factor amplitude toward the unwinded result. Accidental shallow minima can appear for very small shells due to broad momentum mixing.

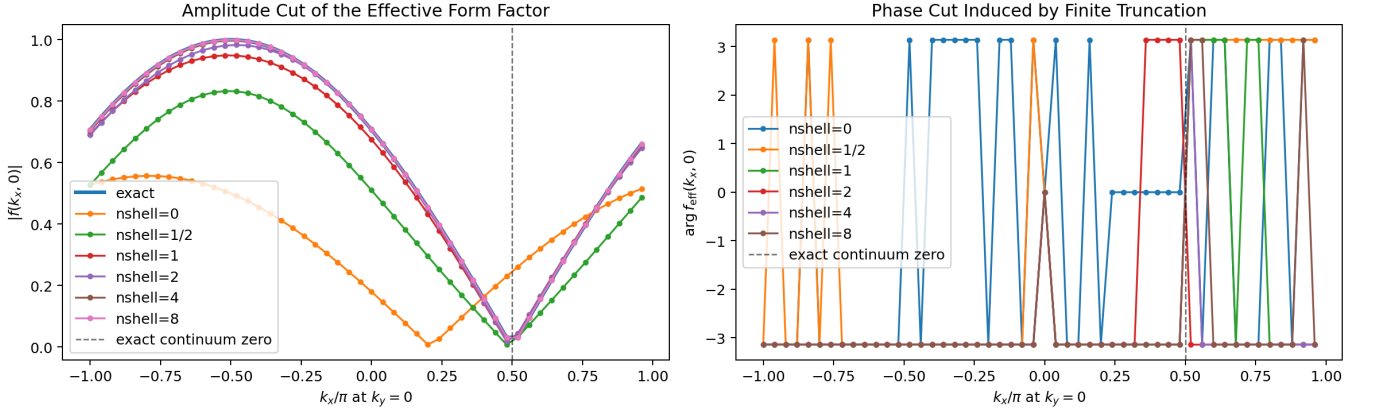


FIG. 6. Cuts at $k_y = 0$ for the effective form factor amplitude (left) and phase (right), with momenta in units of π . The dashed vertical line marks the analytic continuum zero at $k_x/\pi = 0.5$. The $n_{\text{shell}} = 0$ dip near $k_x/\pi \simeq 0.2$ is an accidental minimum, not the original topological zero.

VIII. COMPACT-SUPPORT SPINORS AND THE FHS DIAGNOSTIC

The scalar form factor $f_n^{\text{eff}}(\mathbf{k})$ locates the obstruction encountered when the truncated OW mode is compared with the target lower band. A different object is the normalized two-component trigonometric polynomial

$$|\hat{\psi}_n(\mathbf{k})\rangle = \frac{|\psi_n(\mathbf{k})\rangle}{\|\psi_n(\mathbf{k})\|}, \quad (58)$$

where $|\psi_n(\mathbf{k})\rangle$ is defined in Eq. (57). If $\psi_n(\mathbf{k})$ is nonzero throughout T^2 , then it is itself a smooth periodic global section of the one-dimensional subspace spanned by $\hat{\psi}_n(\mathbf{k})$. Consequently,

$$\psi_n(\mathbf{k}) \neq 0 \quad \forall \mathbf{k} \in T^2 \quad \implies \quad C_{\psi_n} = 0. \quad (59)$$

This elementary global-section argument prevents one from identifying the winding of $\langle u_-(\mathbf{k}) | \psi_n(\mathbf{k}) \rangle$ with a Chern number carried by a nonvanishing compact-support spinor.

For diagnostic purposes one may nevertheless apply the Fukui–Hatsugai–Suzuki (FHS) lattice discretization to $\hat{\psi}_n$:

$$C_{\psi_n}^{\text{FHS}} = \frac{1}{2\pi} \sum_{\mathbf{k}} F_{xy}(\mathbf{k}), \quad (60)$$

where

$$e^{iF_{xy}(\mathbf{k})} = U_x(\mathbf{k}) U_y(\mathbf{k} + \hat{x}) U_x^*(\mathbf{k} + \hat{y}) U_y^*(\mathbf{k}), \quad U_\mu(\mathbf{k}) = \frac{\langle \hat{\psi}_n(\mathbf{k}) | \hat{\psi}_n(\mathbf{k} + \hat{\mu}) \rangle}{|\langle \hat{\psi}_n(\mathbf{k}) | \hat{\psi}_n(\mathbf{k} + \hat{\mu}) \rangle|}. \quad (61)$$

When $\|\psi_n\|$ develops a very sharp but unresolved minimum, a coarse lattice evaluation can cross the admissibility boundary of the discrete formula and report a spurious nonzero integer. Such an integer is not evidence for a nontrivial bundle unless convergence is established while the spinor remains resolved.

A. Results for the square n_{shell} mask

Target lower band. The lower-band projector has nonzero Chern number, equivalently encoded by the overlap-form factor zero at $\mathbf{k}/\pi = (0.5, 0)$. This is the topological object probed by the OW measurement channel.

Finite-support spinor. For the compact-support fields computed from Eq. (57), direct minimization of $\|\psi_n\|$ and the stable FHS evaluation give

n_{shell}	$\min_{\mathbf{k}} \ \psi_n(\mathbf{k})\ $	ν_n of f_n^{eff}	$C_{\psi_n}^{\text{FHS}}$
0	5.571×10^{-1}	−1 (displaced accidental zero)	0
1/2	1.672×10^{-1}	−1	0
1	5.106×10^{-2}	−1	0
2	1.731×10^{-2}	−1	0

(62)

Thus the $n_{\text{shell}} = 1$ overlap zero and winding are retained, while the compact-support spinor itself remains a globally defined trivial section. For $n_{\text{shell}} = 4$ and 8, the minimum norm becomes much smaller (approximately 2.05×10^{-3} and 3.36×10^{-5} , respectively); nonzero FHS values on fixed coarse grids are therefore resolution-sensitive singular-limit diagnostics and are not used here to assign a Chern number to the finite-support spinor.

Berry-flux maps. Figure 7 displays the finite-grid FHS flux diagnostic retained from the numerical study. It visualizes concentration of discrete Berry flux as the compact-support spinor approaches an almost-zero. It should not be read as replacing the overlap-winding test or overriding Eq. (59).

B. Distinction between overlap winding and compact-spinor Chern number

Using the overlap-winding formula

$$\nu_n = \frac{1}{2\pi i} \oint d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln f_n^{\text{eff}}(\mathbf{k}), \quad (63)$$

the contribution of each continuum zero is computed by integrating the phase of f_n^{eff} around a small loop enclosing that zero. In the cases relevant to the circuit, the surviving $\nu_n = -1$ states that the finite-range measurement mode continues to encounter the lower-band Chern obstruction. It does not assert that the standalone trigonometric-polynomial spinor ψ_n is a nontrivial Bloch bundle. This distinction is already visible at $n_{\text{shell}} = 0$, where a constant spinor has $C_{\psi_0} = 0$ although its overlap with a nontrivial target band can possess a winding zero; it remains essential for all finite shells.

IX. SUMMARY

Proposition 1 (Single band). *Let $|\chi_{\mathbf{R}}\rangle$ be a coherent state in a band of Chern number $C \neq 0$, and let $g(\mathbf{r})$ be an arbitrary complex-valued function on the lattice. Define the modulated state $|\Phi^{(g)}\rangle = g(\hat{\mathbf{r}}) |\chi_{\mathbf{R}}\rangle$ and its effective form factor*

$$f^{(g)}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}(\mathbf{k}' - \mathbf{k}) e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}} f(\mathbf{k}) \langle u_{\mathbf{k}'} | u_{\mathbf{k}} \rangle. \quad (64)$$

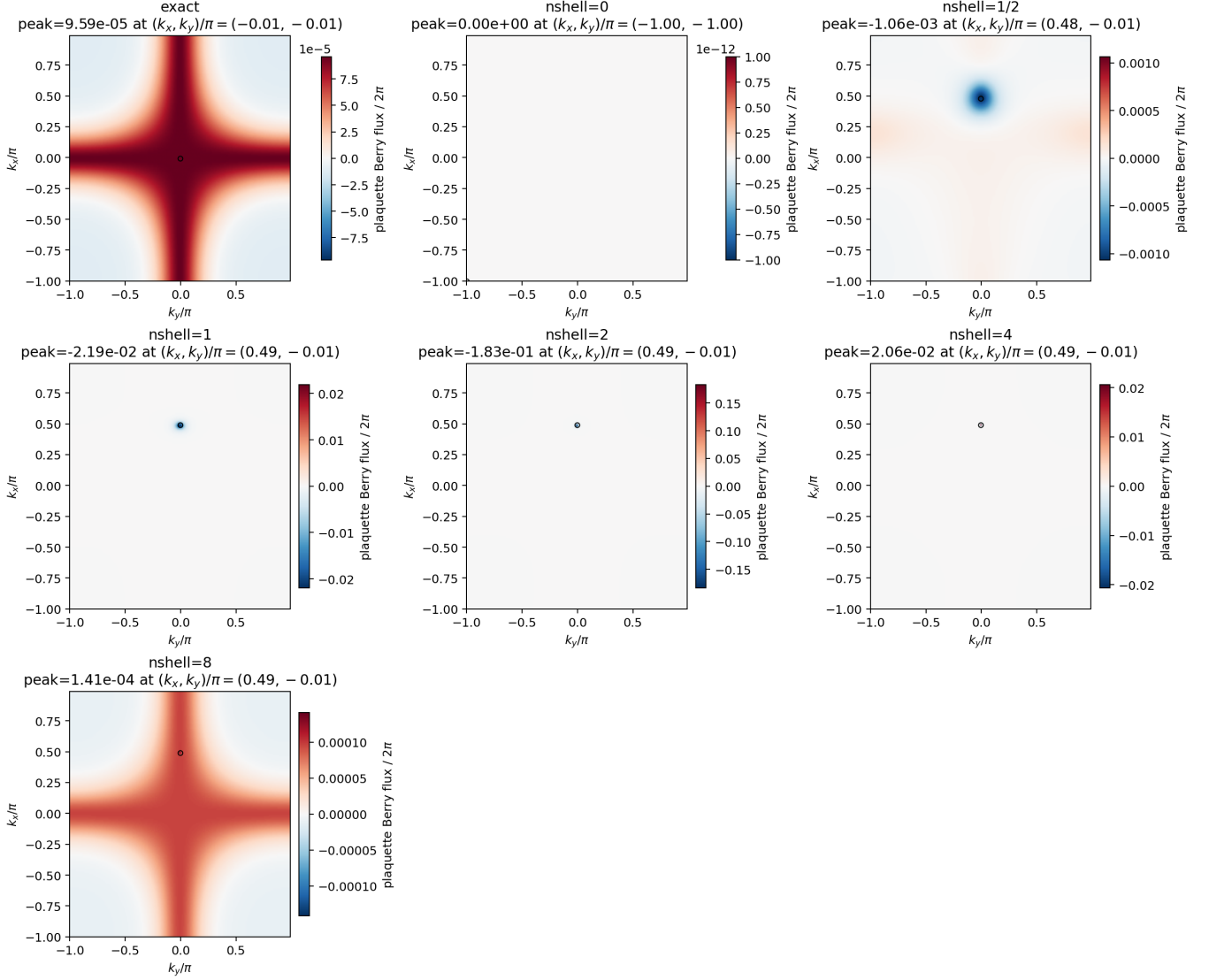


FIG. 7. Finite-grid FHS Berry-flux diagnostic, $F_{xy}(\mathbf{k})/(2\pi)$, for the target lower band and for the compact-support spinor $|\psi_n(\mathbf{k})\rangle$. As the minimum norm of the finite-support spinor becomes small, flux can concentrate within unresolved plaquettes and the discrete integer becomes grid-sensitive. The topological circuit diagnostic used in this note is instead the zero and winding of f_n^{eff} .

Then:

- (i) (**Overlap criterion.**) The overlap-winding index $\nu_g = \frac{1}{2\pi i} \oint d\mathbf{k} \cdot \nabla_{\mathbf{k}} \ln f^{(g)}(\mathbf{k})$ vanishes whenever $f^{(g)}$ is nowhere vanishing on T^2 .
- (ii) (**Preservation.**) If $\tilde{g}(\mathbf{q})$ produces a sufficiently small perturbation on a loop surrounding an unwinded zero, then $f^{(g)}$ inherits that zero's winding and $\nu_g = -C$.
- (iii) (**Destruction.**) Sufficiently broad momentum mixing can remove the original overlap zero; if no replacement zero is generated, $\nu_g = 0$.
- (iv) (**Box window.**) For $\sigma \gg \xi_{\text{rms}}$, the narrow-kernel/Rouché argument protects overlap winding while allowing displacement of the zero.
- (v) (**Length scales.**) The quantum-geometric inequality constrains $\xi_{\text{rms}} \geq \xi_{\text{top}} = \sqrt{2|C|} \ell$, whereas ξ_{exp} controls only the asymptotic real-space tail.

(vi) (**Compact spinor.**) If the finite-support spinor $\psi_n(\mathbf{k})$ is smooth, periodic, and nowhere zero, then it is a global section and its own Chern number is zero, independently of ν_n .

Proposition 2 (Multiband). Let $|\chi_{m,\mathbf{R}}\rangle$, $m = 1, \dots, M$, be coherent states spanning an M -band subspace with total Chern number $C \neq 0$. Define the effective overlap matrix $F_{nm}^{(g)}(\mathbf{k}')$ by (32). Then:

- (i') (**Overlap criterion.**) The determinant-overlap winding defined in Eq. (33) vanishes when $\det F^{(g)}$ is nowhere zero.
- (ii') (**Preservation.**) A sufficiently gentle convolution preserves the determinant-zero winding of the target subspace.
- (iii') (**Interpretation.**) Without an in-subspace frame condition, $\nu_g^{(M)}$ is an overlap diagnostic and must not be automatically identified with a Chern invariant of the modulated vectors.
- (iv') (**Critical scale.**) The parametric window scale is set by ξ_{rms} , with $\xi_{\text{rms}} \geq \sqrt{2|C|}\ell$.

The key structural difference from the single-band case is that $\det F^{(g)}$ is a polynomial in the convolved matrix elements, so zeros can be created or annihilated in pairs at intermediate scales.

X. PHYSICAL INTERPRETATION AND CIRCUIT IMPLICATIONS

A. Topology stored in zero structure

The Wannier obstruction of the target band is stored in the zeros of $f(\mathbf{k})$, which are phase singularities carrying its overlap winding. Any real-space window acts as a momentum-space twisted convolution and may shift, create, or remove zeros of the overlap with that target band. The RMS spread of the unwrapped OW mode is subject to a quantum-geometric lower bound, but neither that bound nor the exponential tail length alone determines the fate of a given discrete mask. For the square mask the fate of the obstruction is resolved directly by the continuum form-factor cut.

B. Universality of the criterion

The general formula (7) makes clear that the overlap-zero structure depends on both the momentum-space width and the symmetry of $\tilde{g}$, as well as on the Bloch-overlap dressing. Width compared with $1/\xi_{\text{rms}}$ provides a controlled parametric guide for gentle windows; the explicit square-mask calculation is the relevant determination at small integer n_{shell} .

C. Implications for the adaptive Markov circuit

In the adaptive circuit studied in the companion code, the OW measurement modes are constructed with a finite n_{shell} cutoff: the Wannier mode centered at site $\mathbf{R}$ is restricted to a $(2n+1) \times (2n+1)$ real-space patch. The analysis here shows that this truncation:

1. For $n_{\text{shell}} = 1$, the effective form factor $f_n^{\text{eff}}(\mathbf{k})$ has a zero with the inherited winding, displaced only slightly from the exact continuum location $\mathbf{k}/\pi = (0.5, 0)$. It is therefore a compact finite-range measurement channel which still probes the Chern-band overlap obstruction.
2. At $\alpha = 1$, the same 3×3 window retains 0.992155 of the untruncated OW norm. The longer value $\xi_{\text{exp}} \simeq 1.443a$ characterizes its residual tail and does not force replacement of this channel by $n_{\text{shell}} = 2$.
3. The finite-support spinor should not be assigned the Chern number of the target lower band merely from this overlap zero. For the stable $n_{\text{shell}} = 1$ compact-spinor FHS check, $C_{\psi_1}^{\text{FHS}} = 0$, as required for a nonvanishing global section.

The special case $n_{\text{shell}} = 0$ is qualitatively less faithful: its overlap zero is strongly displaced and arises even though its compact spinor is momentum independent. For quantitative long-time observables, including the particle-transfer Lyapunov gap, comparison of $n_{\text{shell}} = 1$ against $n_{\text{shell}} = 2$ remains a useful convergence test; it is not required to establish survival of the form-factor winding.

The geometric scale $\xi_{\text{top}} = a/\sqrt{\pi} \simeq 0.564a$ gives a lower bound on the RMS size of an OW mode in the $\alpha = 1$, $|C| = 1$ band. It is not a certification radius for a discrete truncation. The justification for $n_{\text{shell}} = 1$ is instead the direct combination of high retained norm and the explicitly preserved overlap zero.

ACKNOWLEDGMENTS

This working note accompanies the adaptive class-A Markov circuit project. The analysis builds on the overcomplete Wannier framework of Li, Dong, Ledwith, and Khalaf [1].

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- [1] Q. Li, J. Dong, P. J. Ledwith, and E. Khalaf, arXiv:2407.02561 (2024).
 - [2] T. Fukui, Y. Hatsugai, and H. Suzuki, J. Phys. Soc. Jpn. **74**, 1674 (2005).